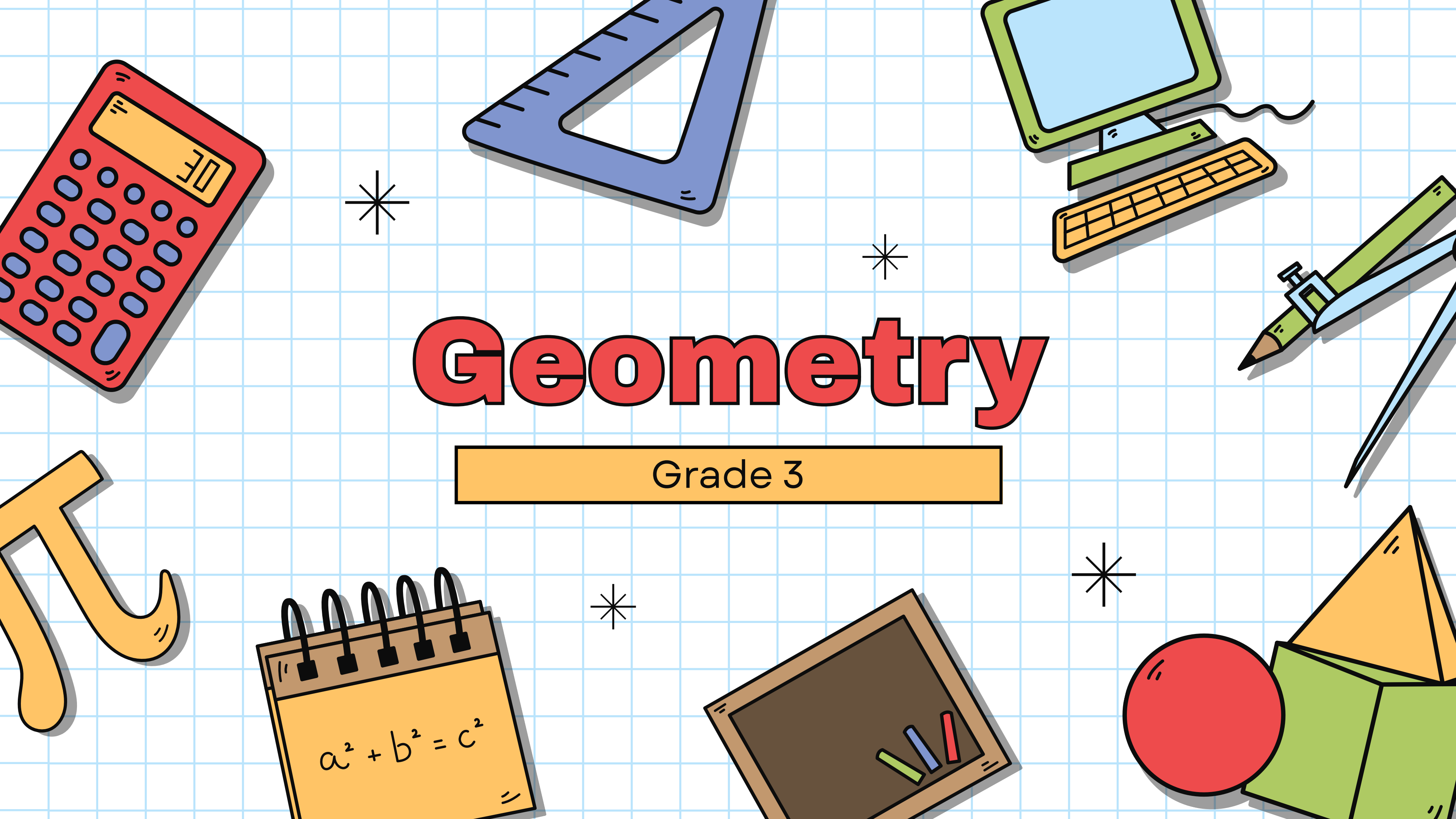




Geometry

Grade 3

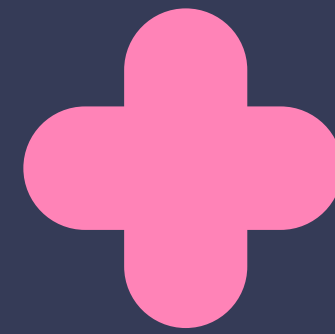
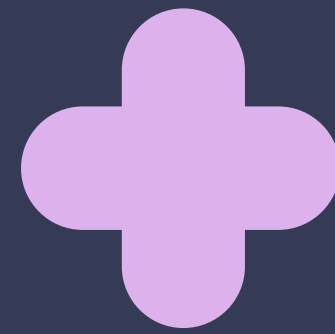
WHAT IS GEOMETRY?





DEFINITION

- Geometry is a branch of mathematics that studies shapes, sizes, angles, and dimensions of objects.
 - It helps us understand the world around us, from the structure of buildings to the patterns in nature.
- 
- 
- 

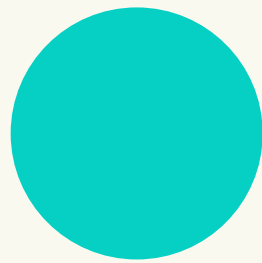


BASIC CONCEPTS IN GEOMETRY

Introduction to fundamental elements in geometry.



POINTS, LINES, AND PLANES



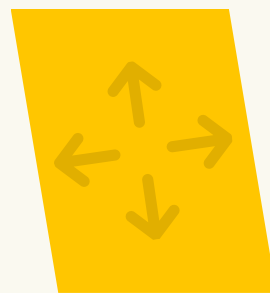
Point

A location in space with no size (denoted by a dot, like A).



Line

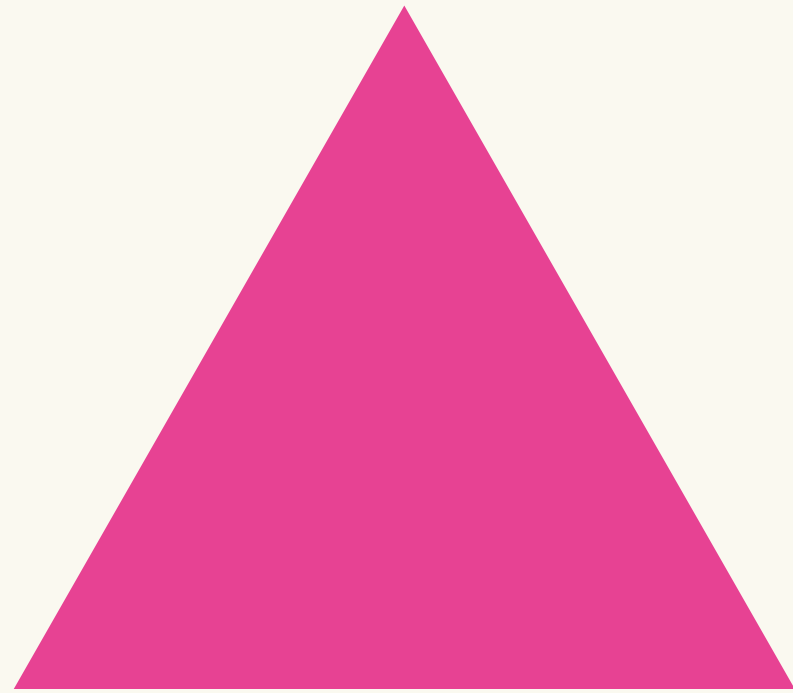
A straight path that extends infinitely in both directions.



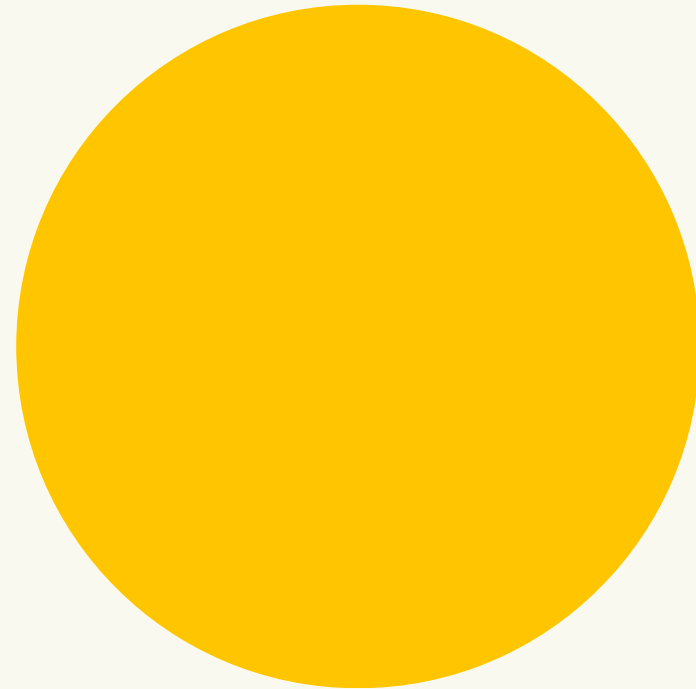
Plane

A flat surface that extends infinitely in all directions.

2D SHAPES (FLAT SHAPES)



Triangle (3 sides)

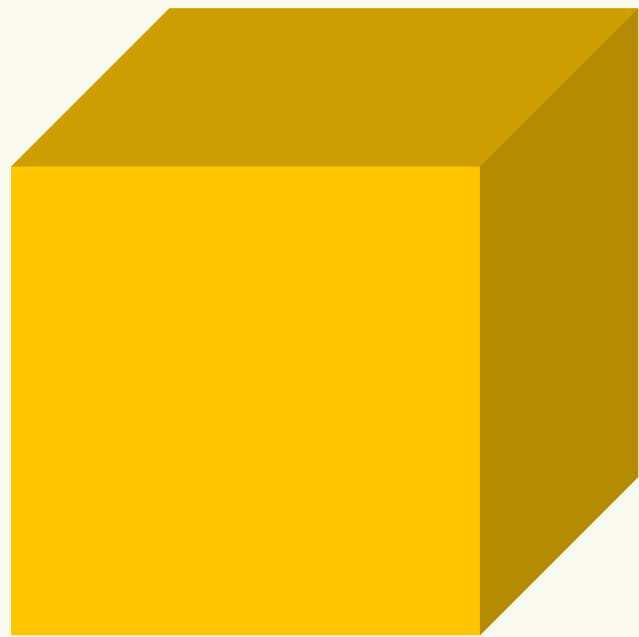


Circle (A round shape with no sides)

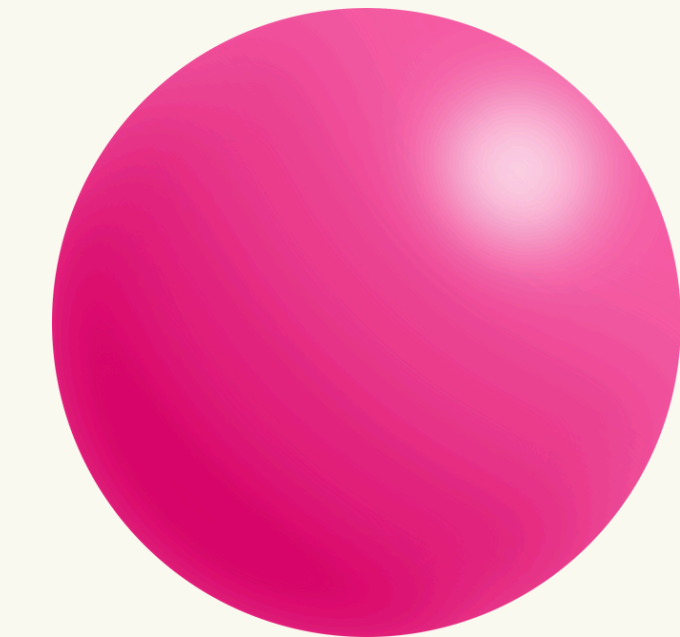
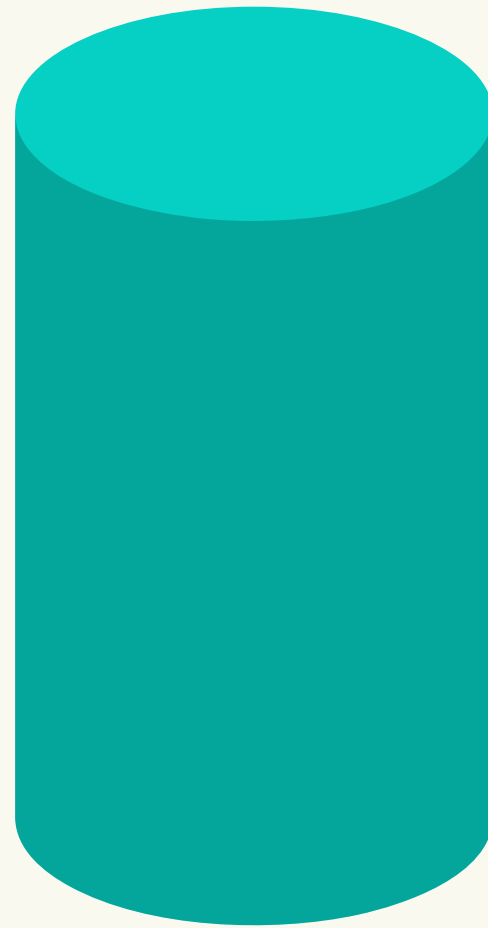


Quadrilateral (4 sides)

3D SHAPES (SOLID SHAPES)

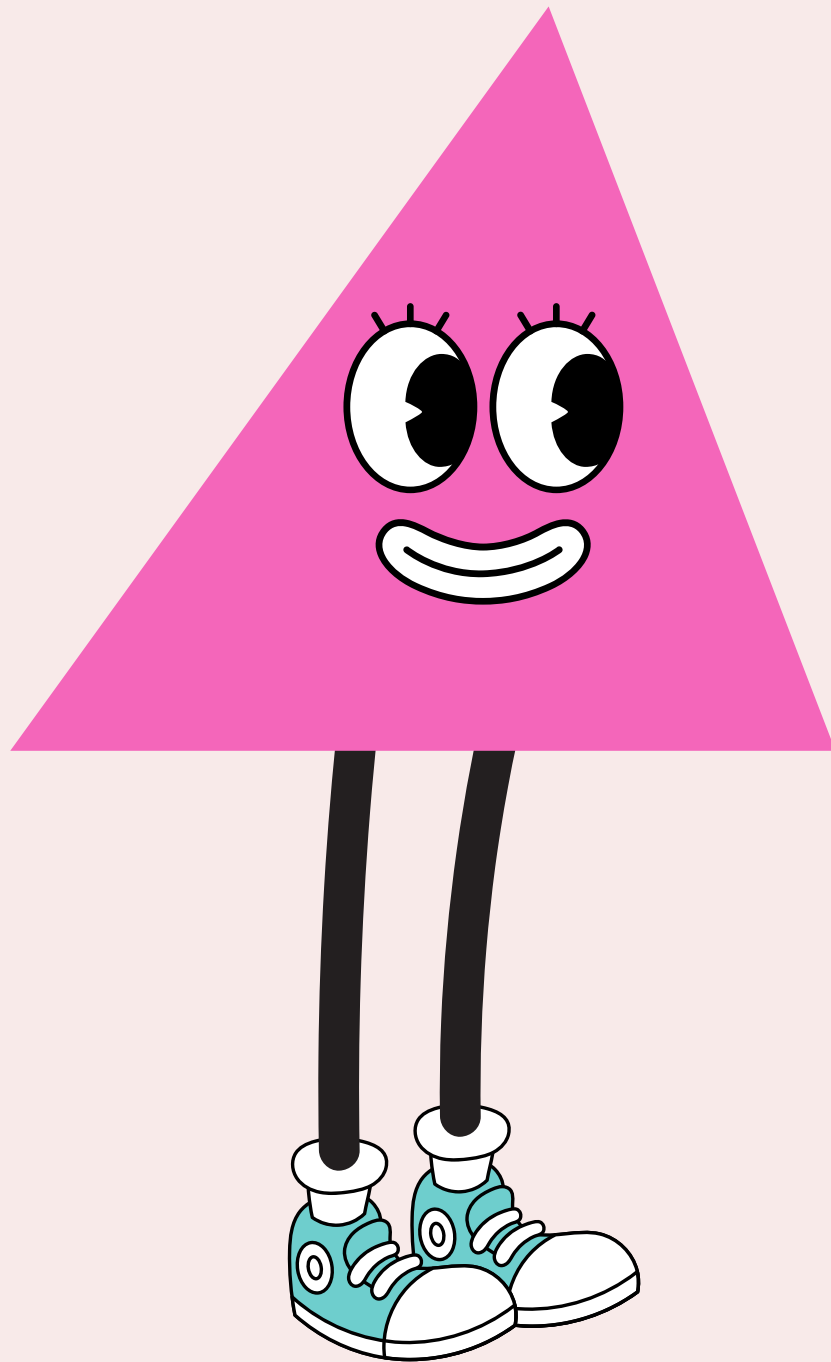


Cube (6 square faces)



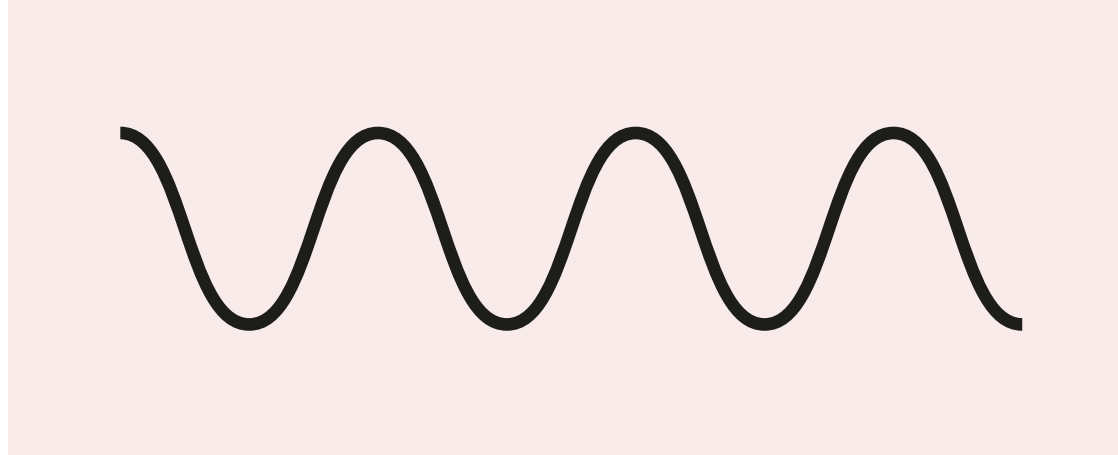
Sphere (Perfectly round)

Cylinder (Circular base
with a curved surface)

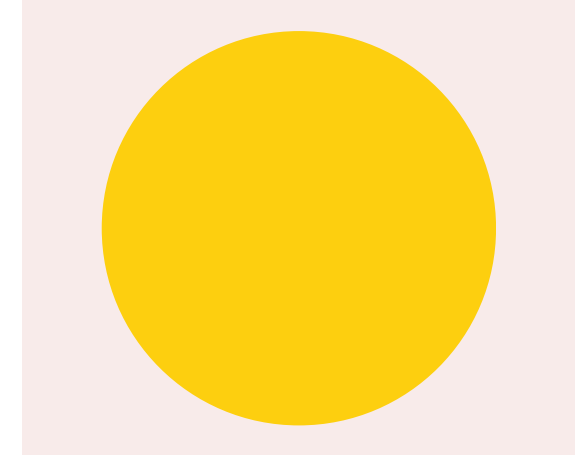


WHAT IS A POLYGON?

- A polygon is a geometric figure made up of straight sides.
- There are no curved lines or open shapes.



wavy line

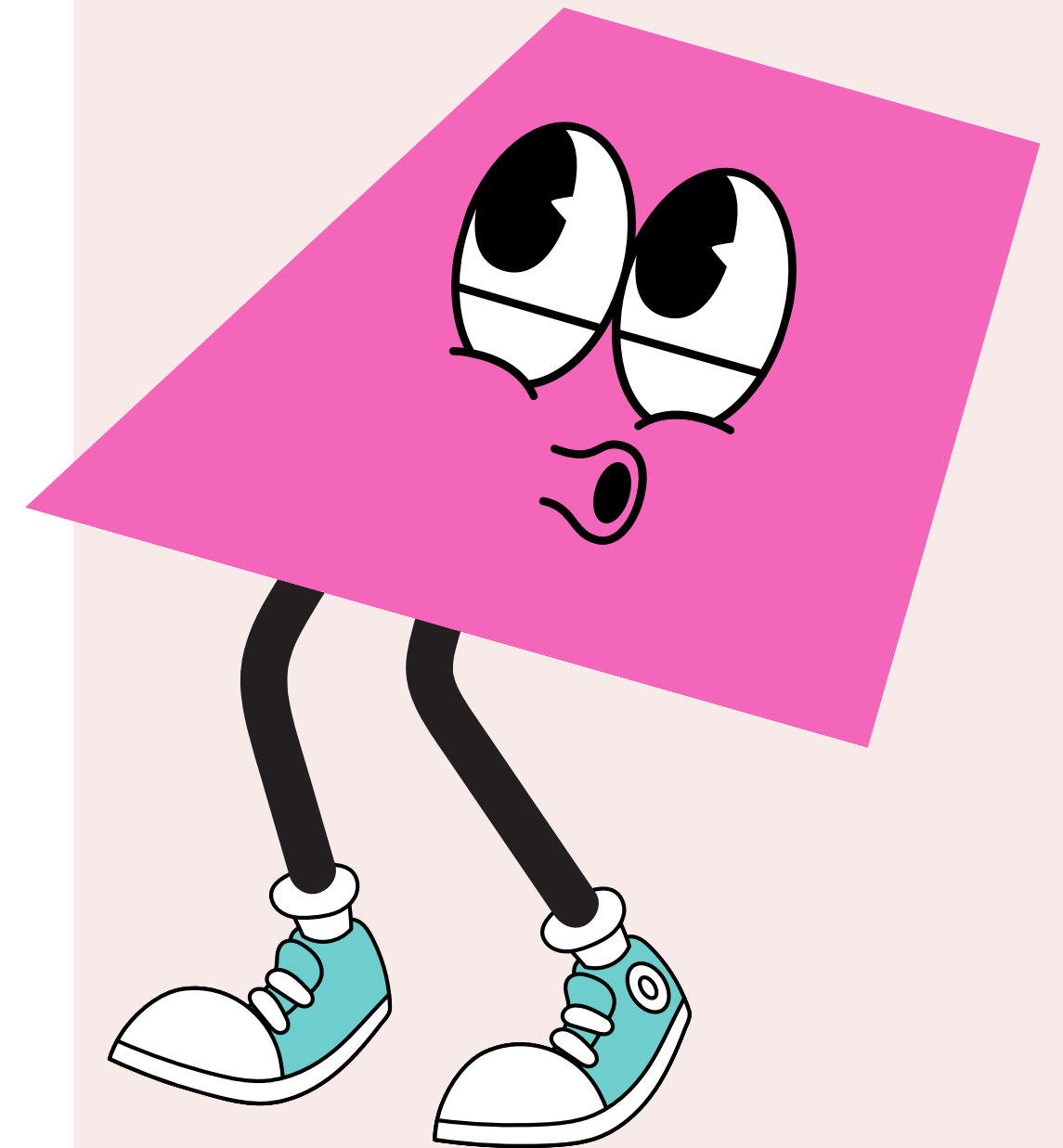
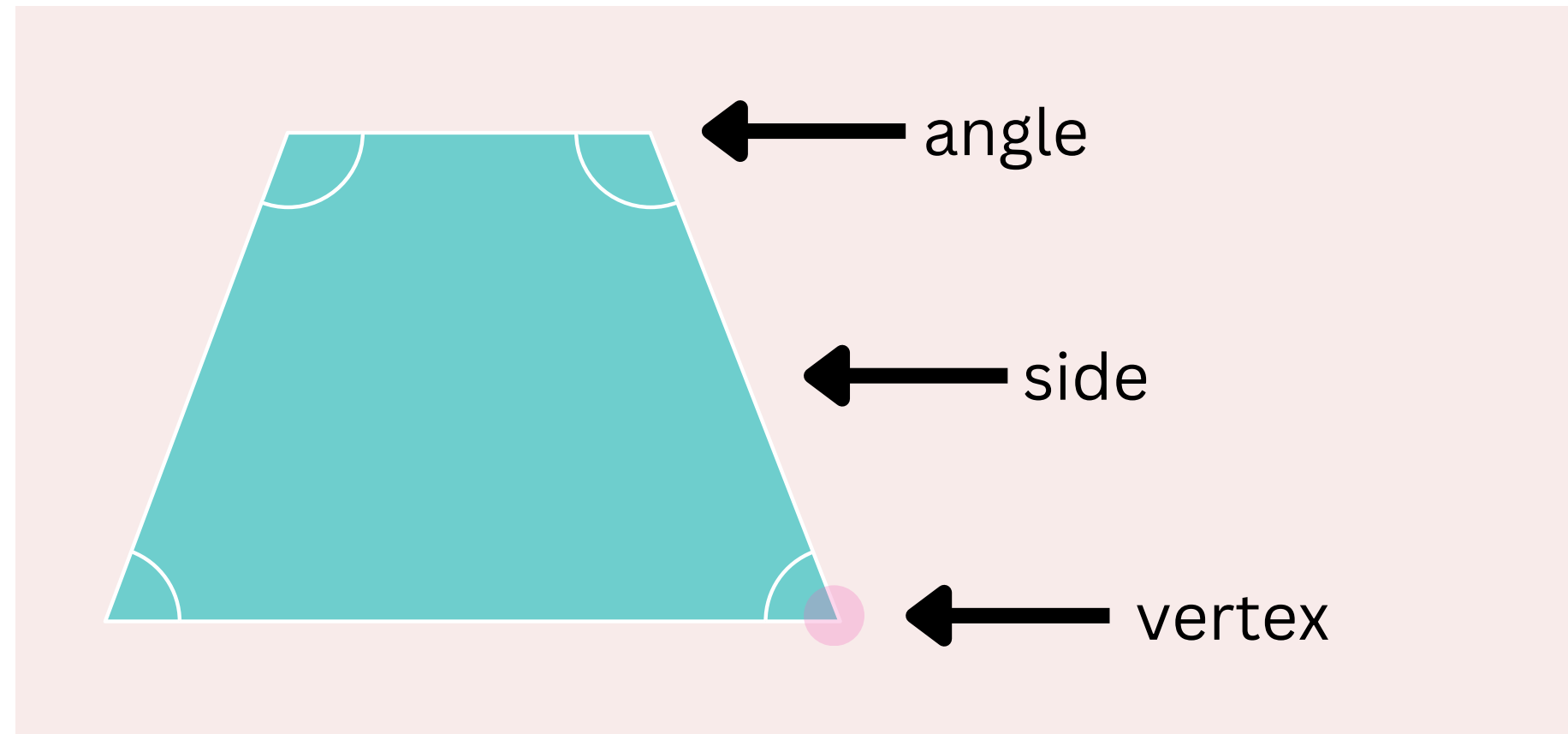


circle

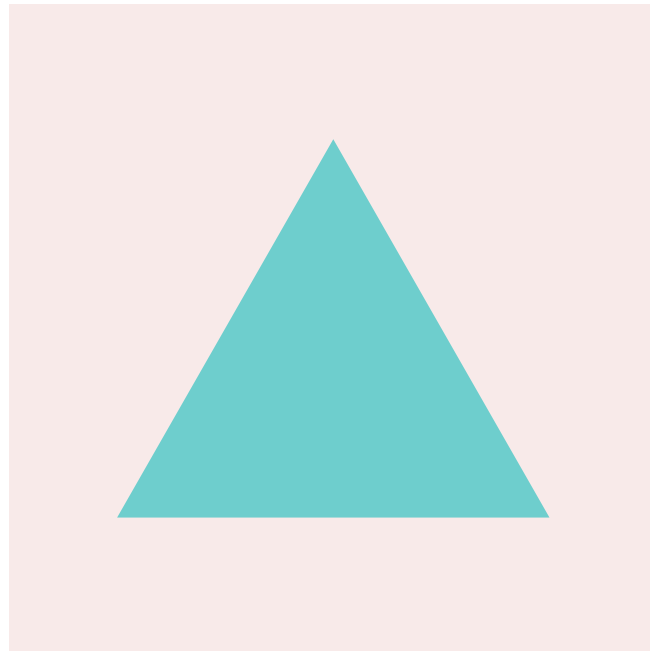
Are there any polygons in the pictures above?

FEATURES OF POLYGONS

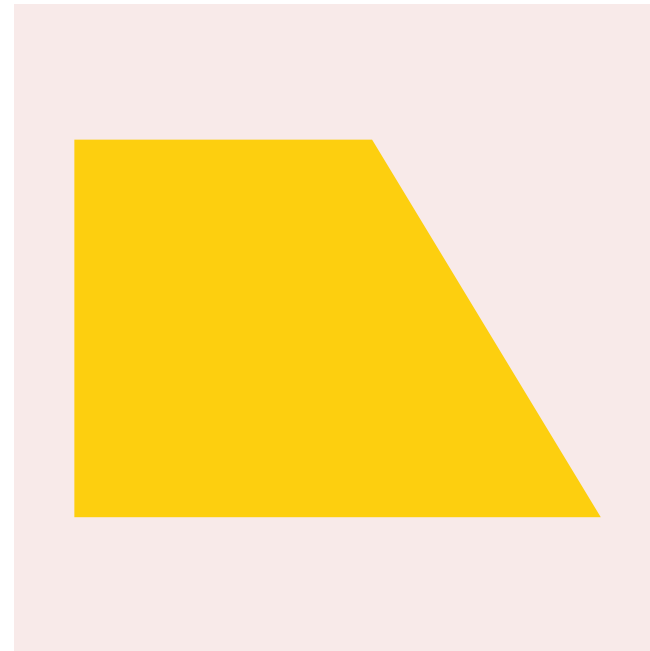
- **Sides** – edges of the figure.
- **Angles** – where the sides meet.
- **Vertices** – points where sides intersect.



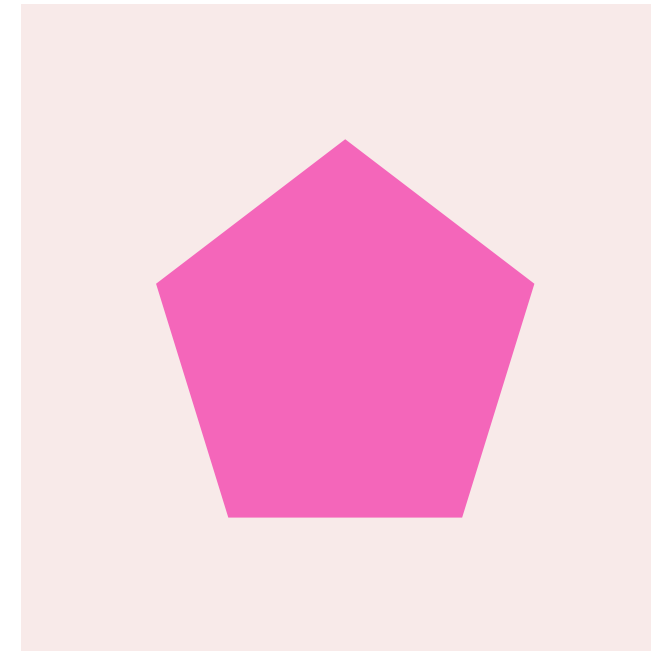
NUMBER OF SIDES AND NAMES OF POLYGONS



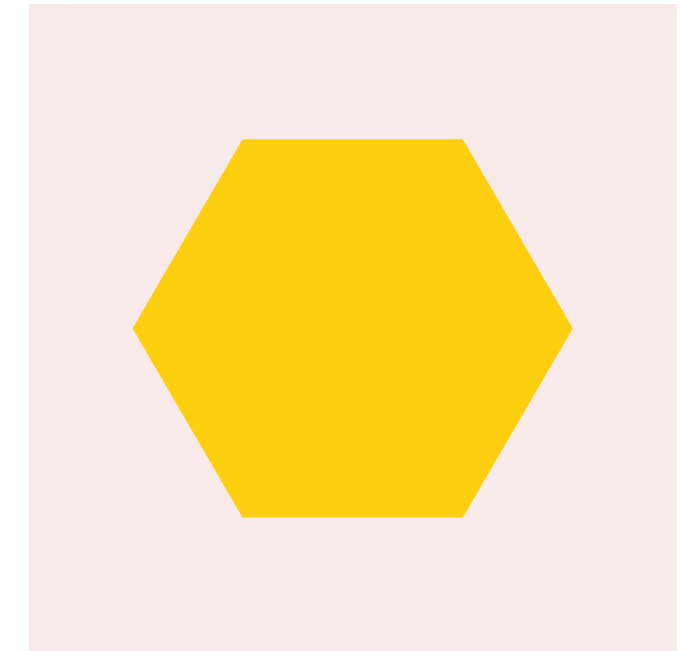
Triangle



Quadrilateral



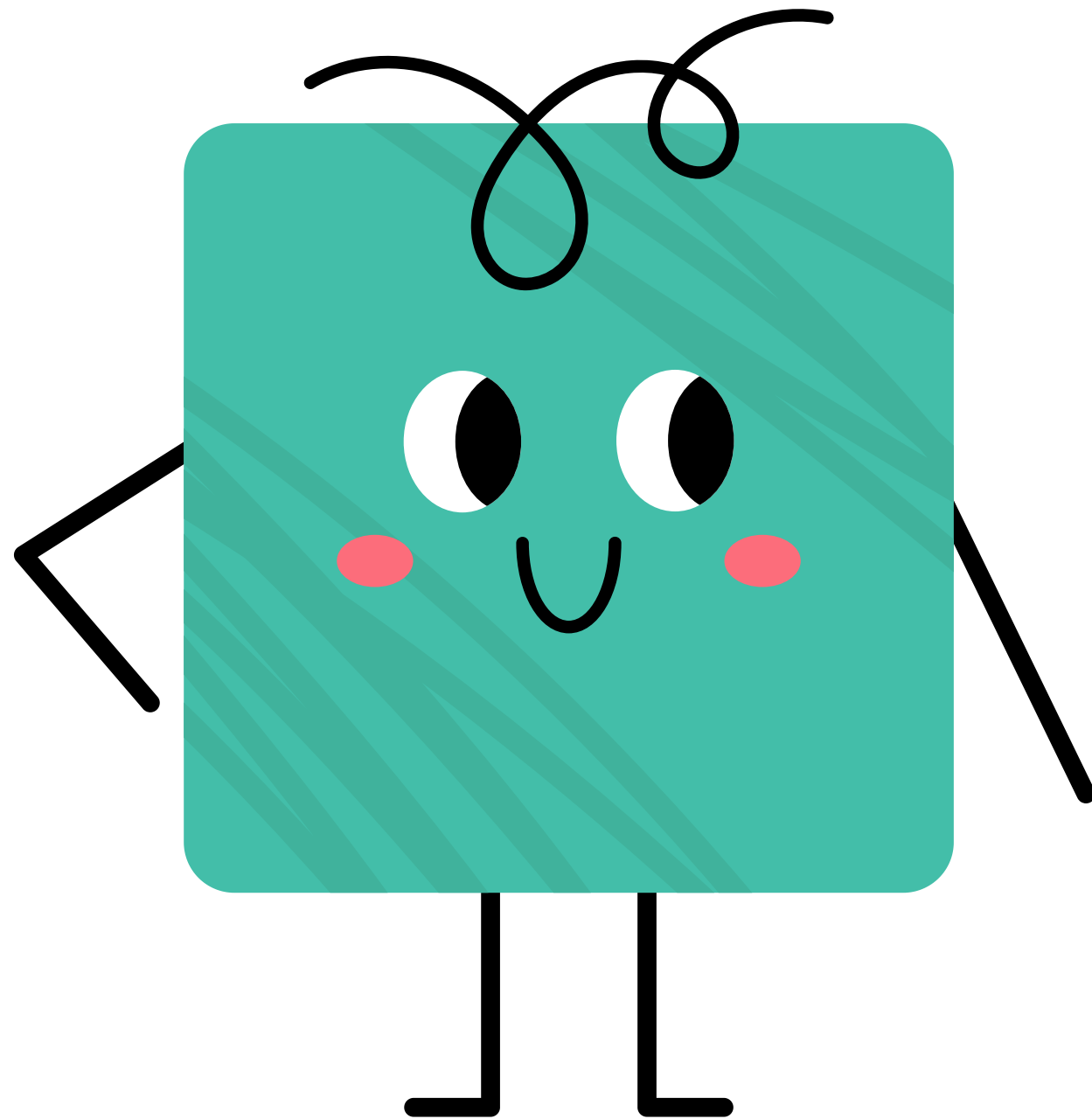
Pentagon



Hexagon

Count the sides and read the names. Does everything match?

SQUARE



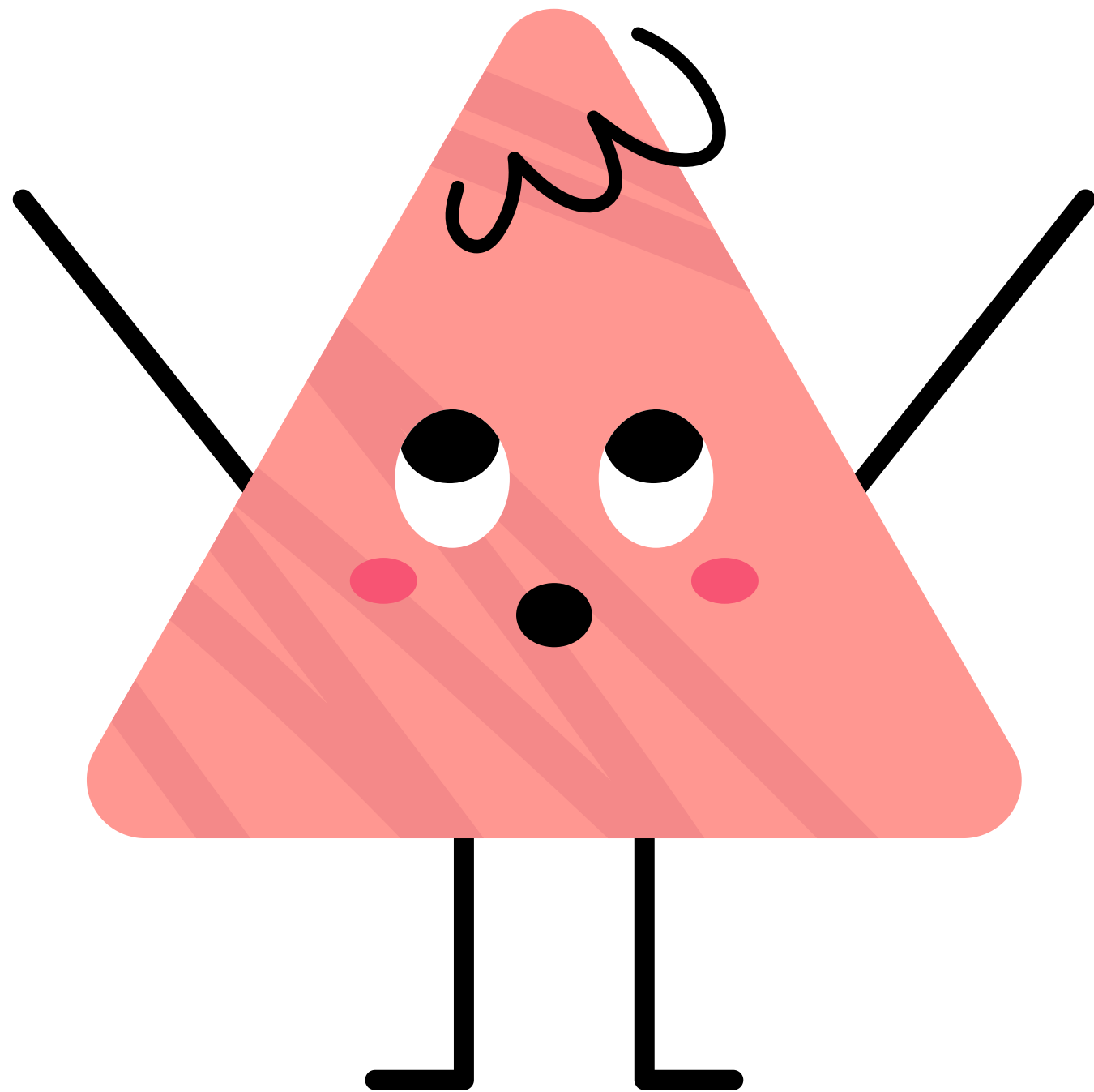
A square is a four-sided shape. All of its sides are exactly the same length.

Sides: 4

Corners: 4

Edges: 4

TRIANGLE



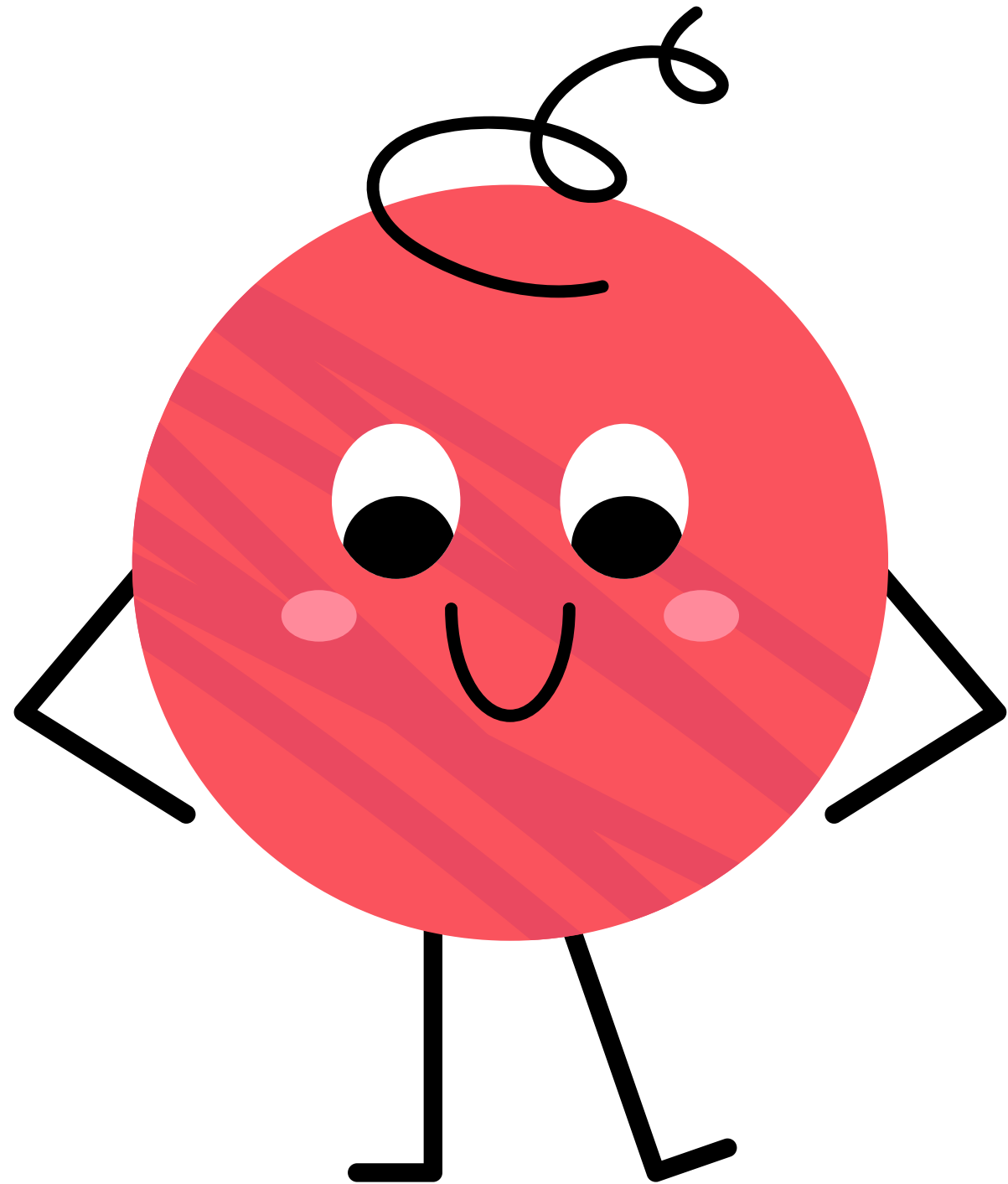
A triangle is a three-sided shape. There are many different types of triangles including equilateral, isosceles and scalene.

Sides: 3

Corners: 3

Edges: 3

CIRCLE



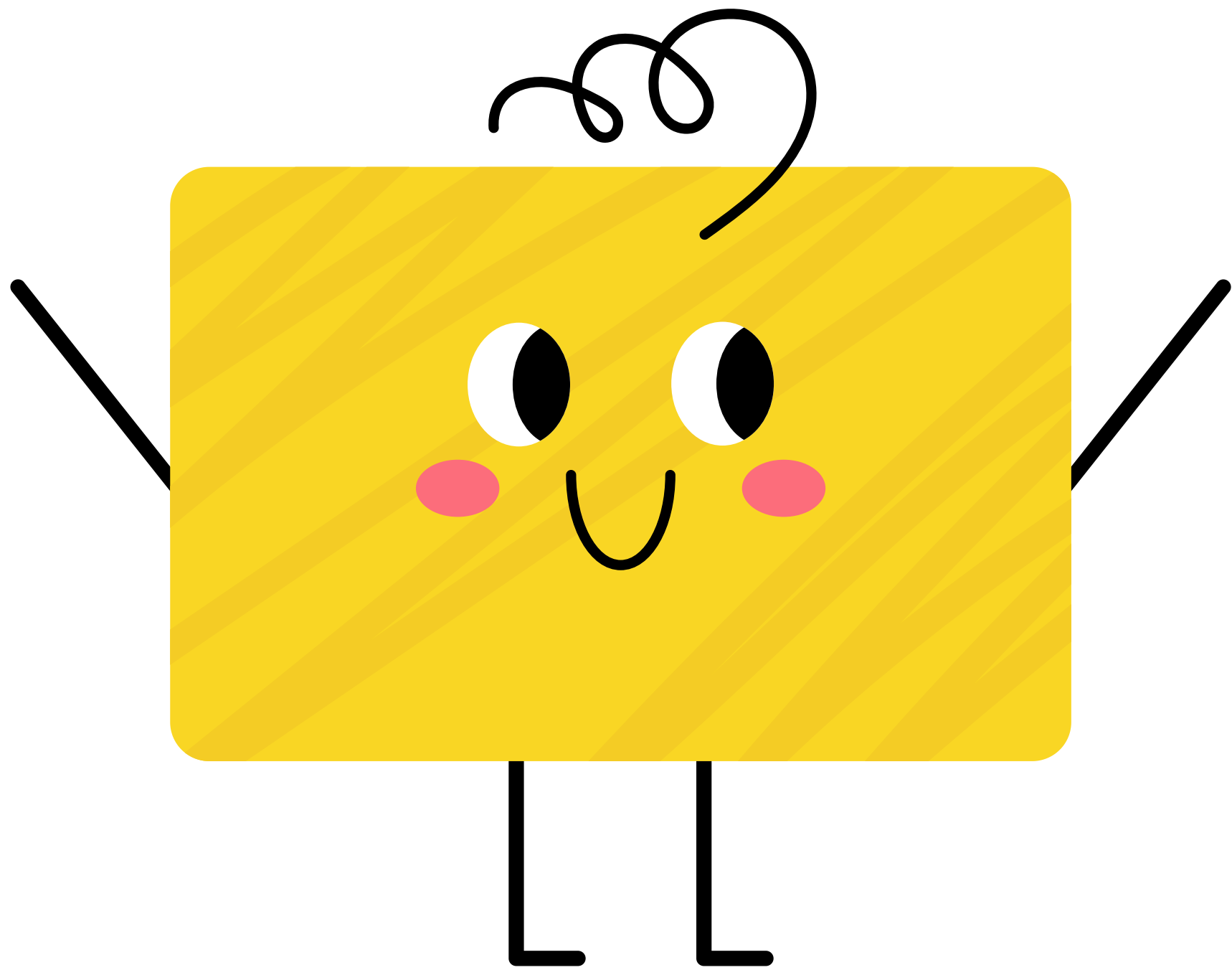
A circle is a round, flat shape. It has one curved side.

Sides: 1

Corners: 0

Edges: 0

RECTANGLE



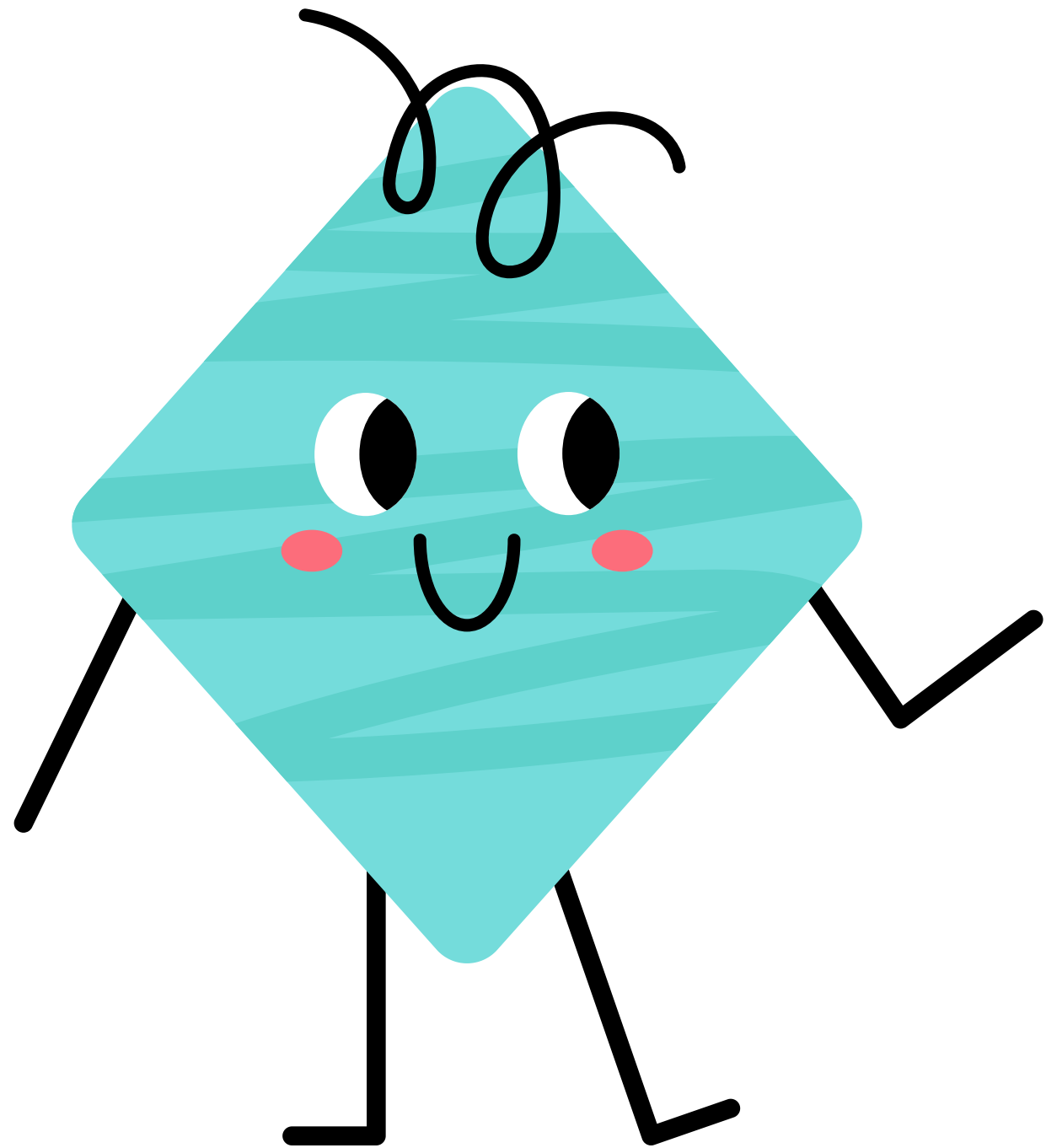
A rectangle is a four-sided shape. Its opposite sides are equal lengths.

Sides: 4

Corners: 4

Edges: 4

RHOMBUS



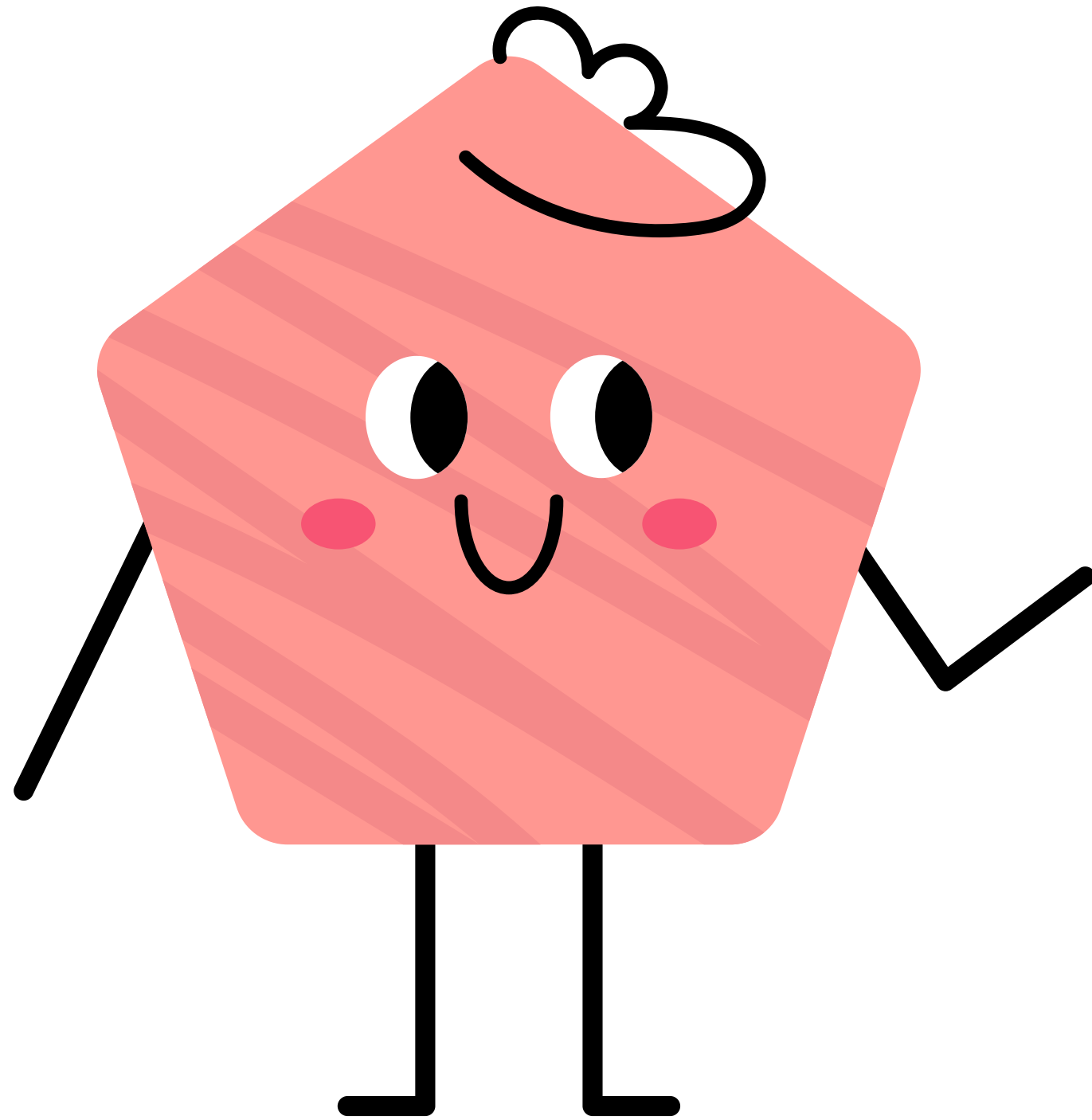
A rhombus is a four-sided shape. It is like a leaning square.

Sides: 4

Corners: 4

Edges: 4

PENTAGON



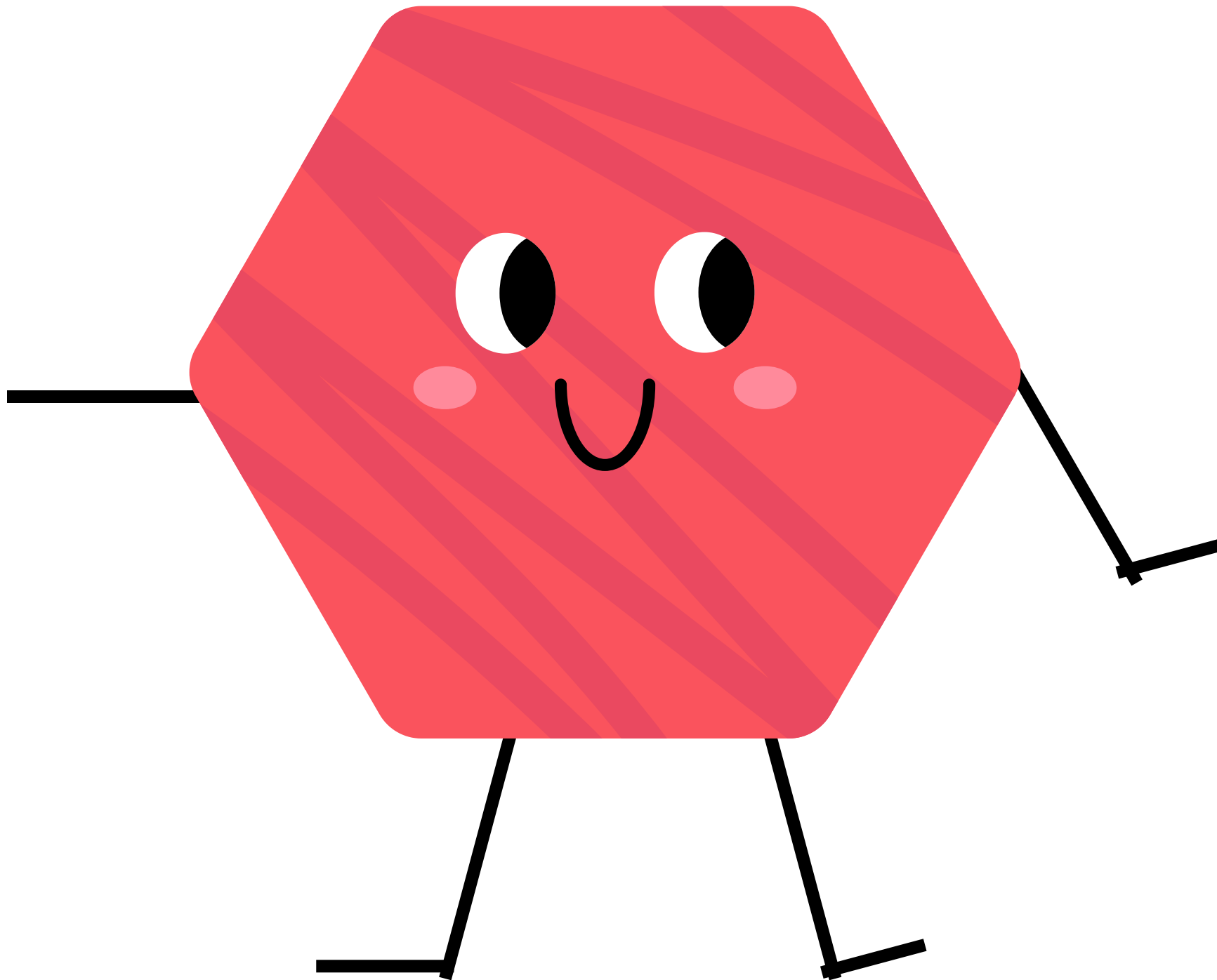
A pentagon is a five-sided shape.

Sides: 5

Corners: 5

Edges: 5

HEXAGON



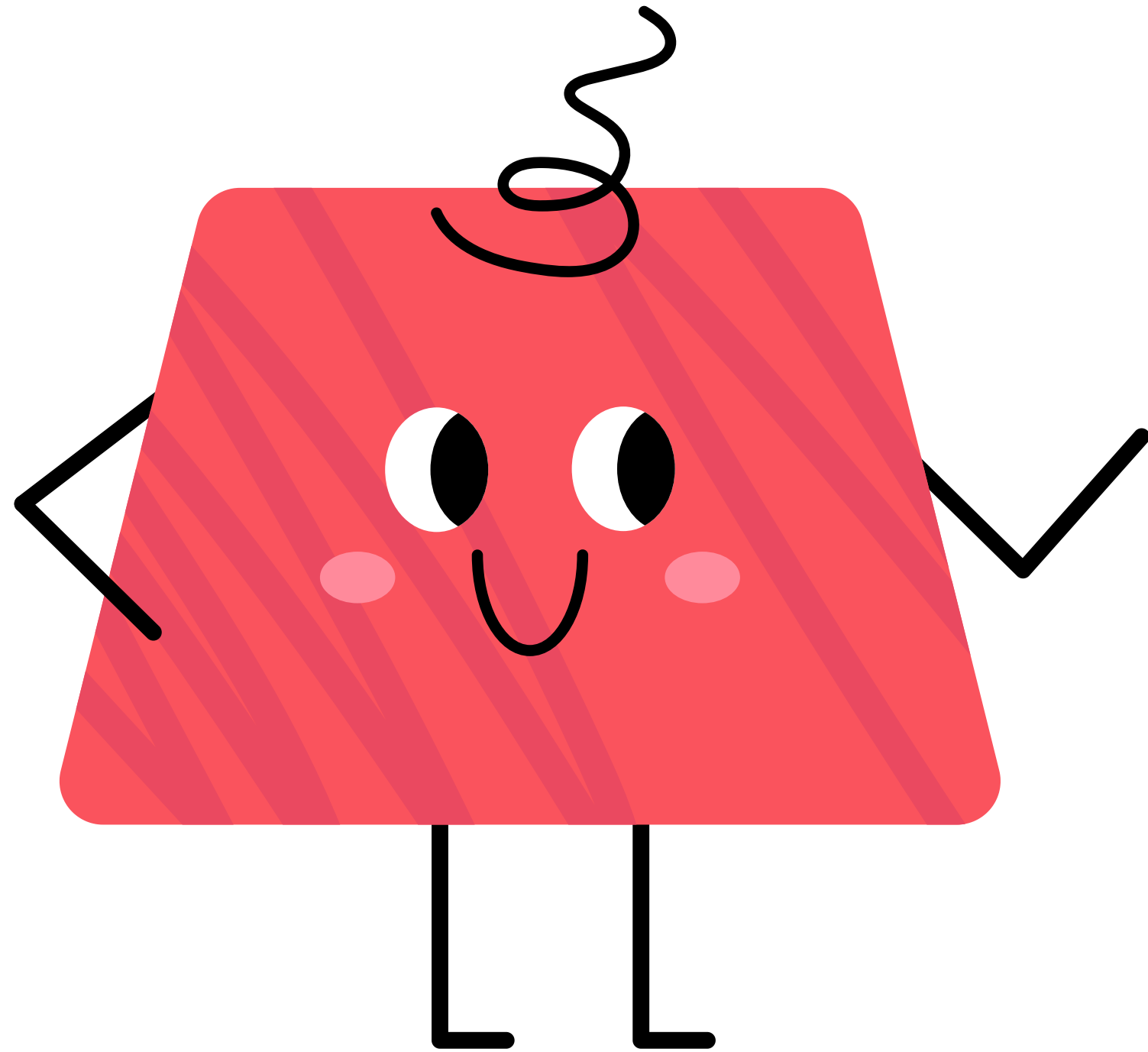
A regular hexagon has six equal sides.

Sides: 6

Corners: 6

Edges: 6

TRAPEZIUM



A trapezium is a four-sided shape that has one set of parallel sides.

Sides: 4

Corners: 4

Edges: 4

OVAL



An oval is a round, flat shape. It has one curved side.

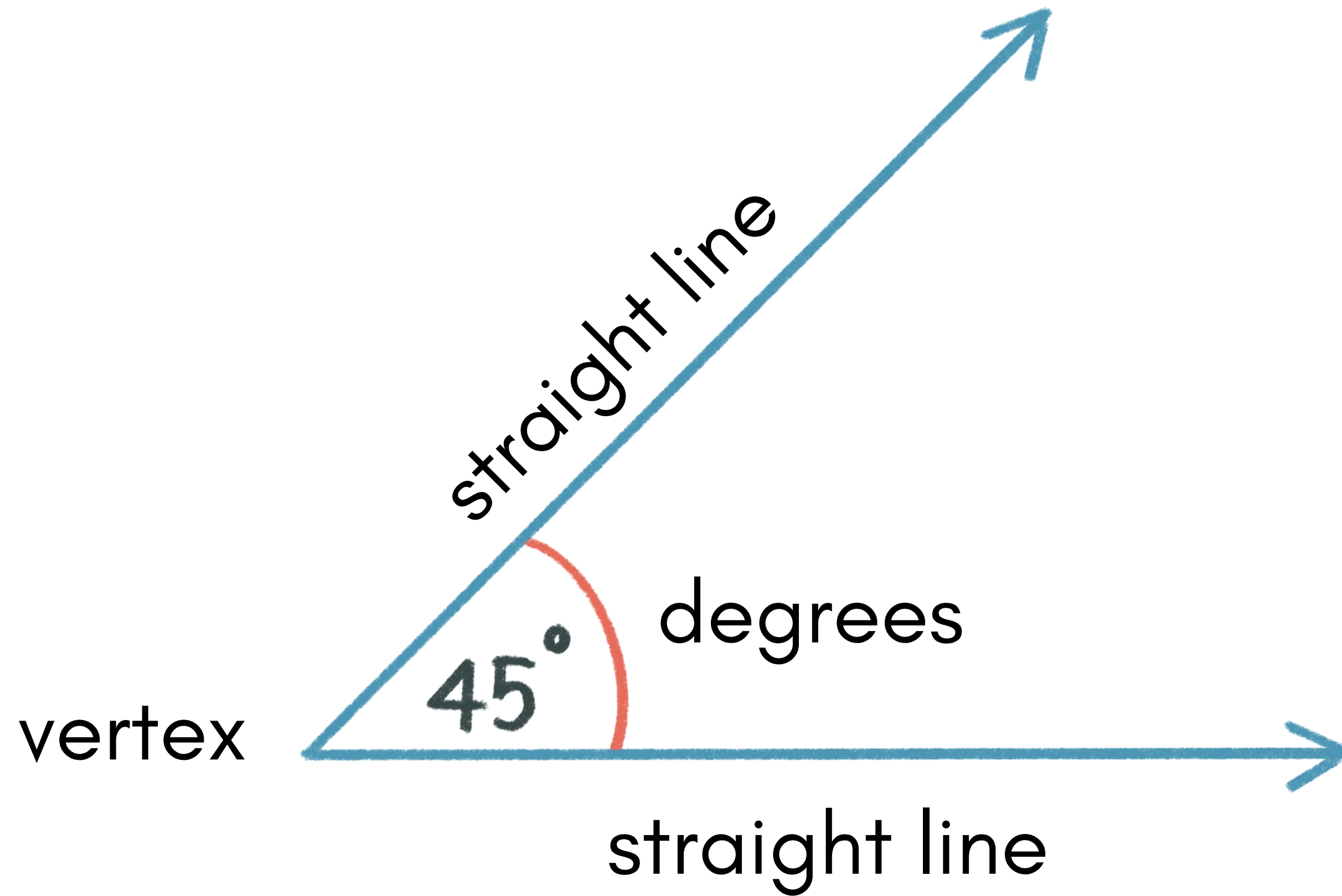
Sides: 1

Corners: 0

Edges: 0

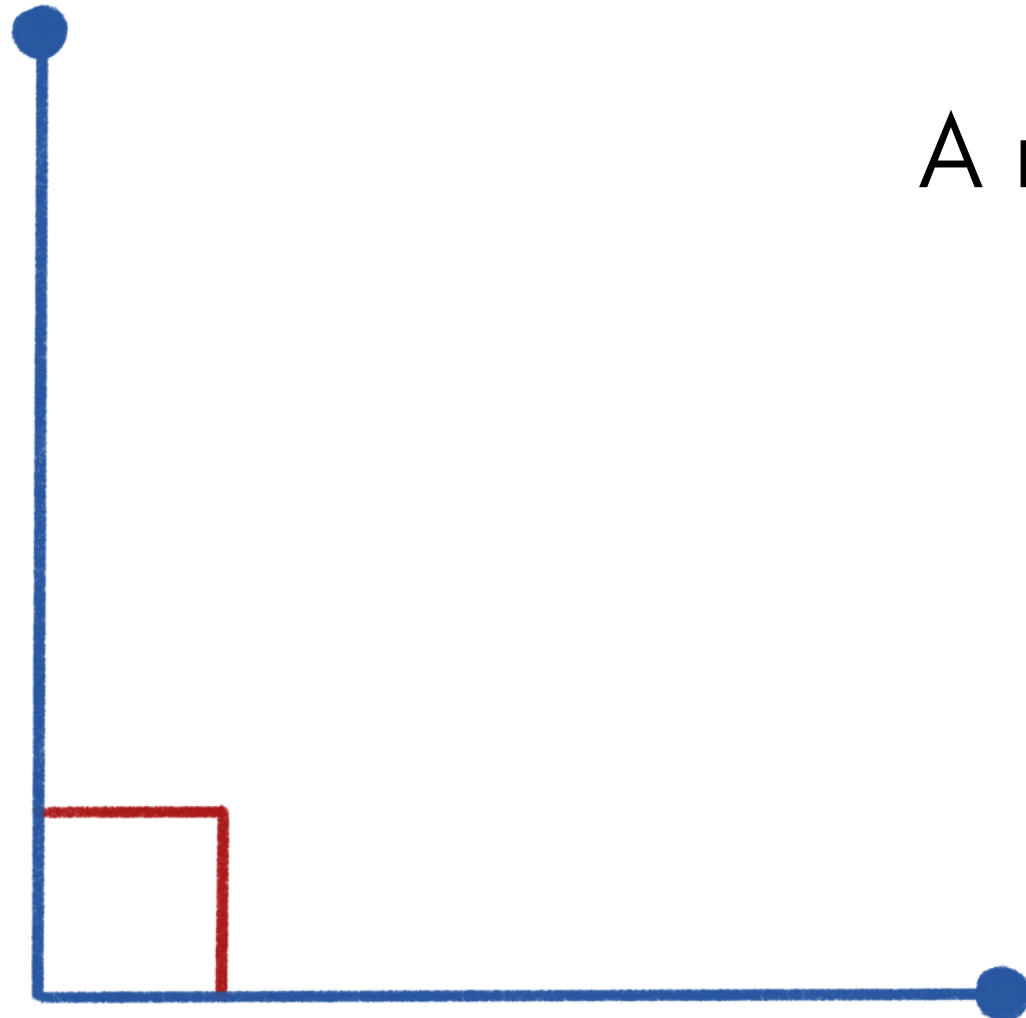
WHAT IS AN ANGLE?

An angle is the shape formed by two rays meeting at a common end point.

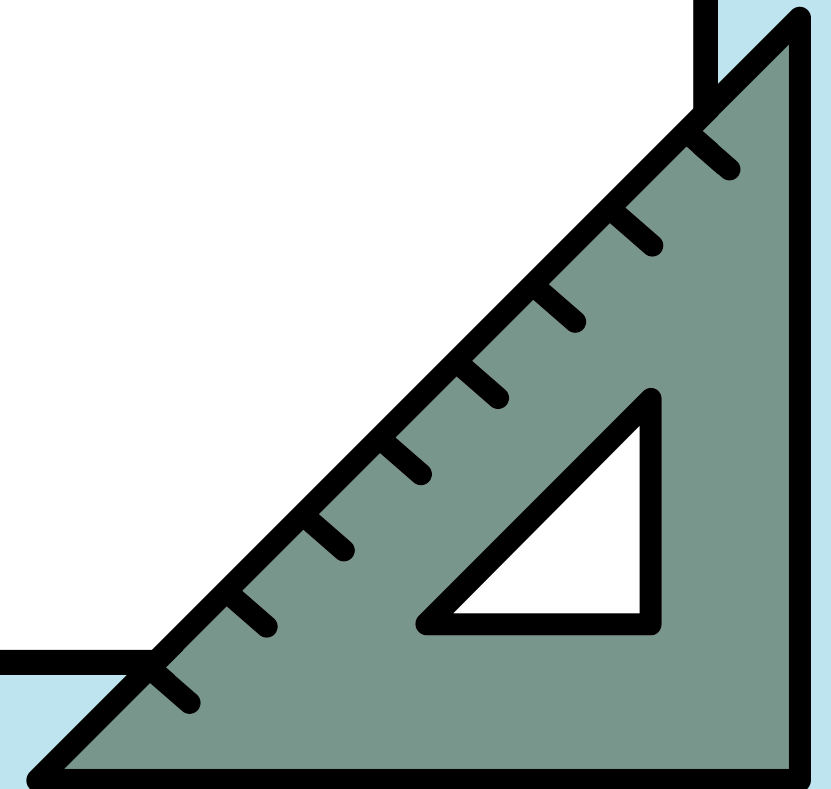
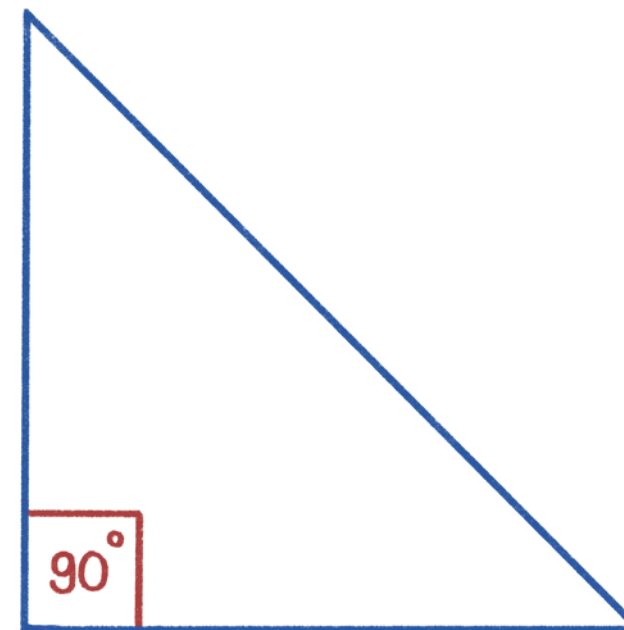


RIGHT ANGLES

A right angle is equal to 90°

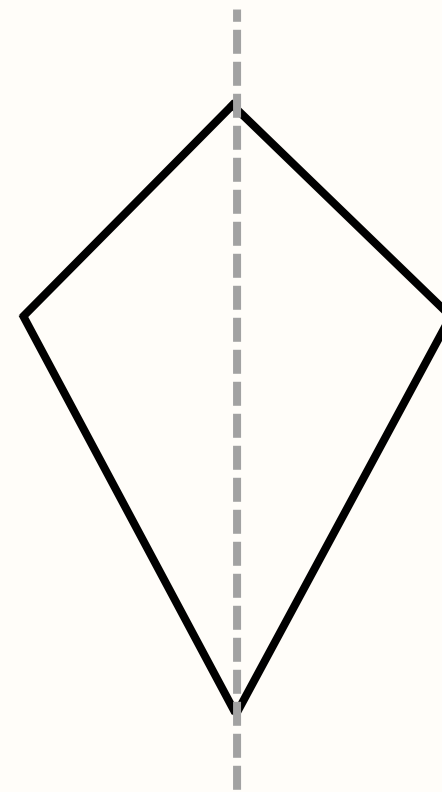
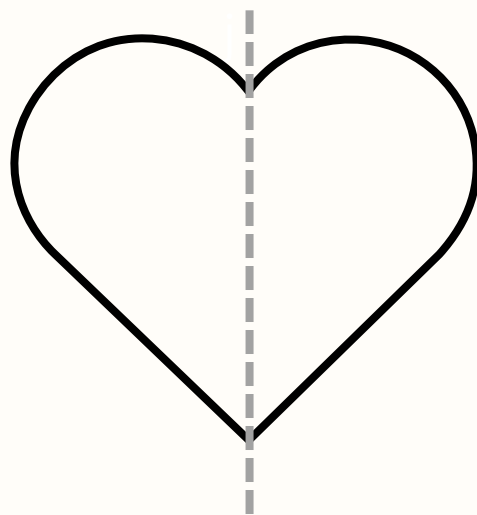
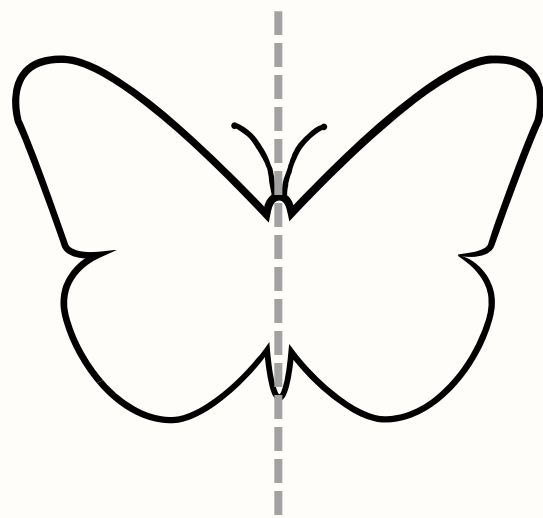


right



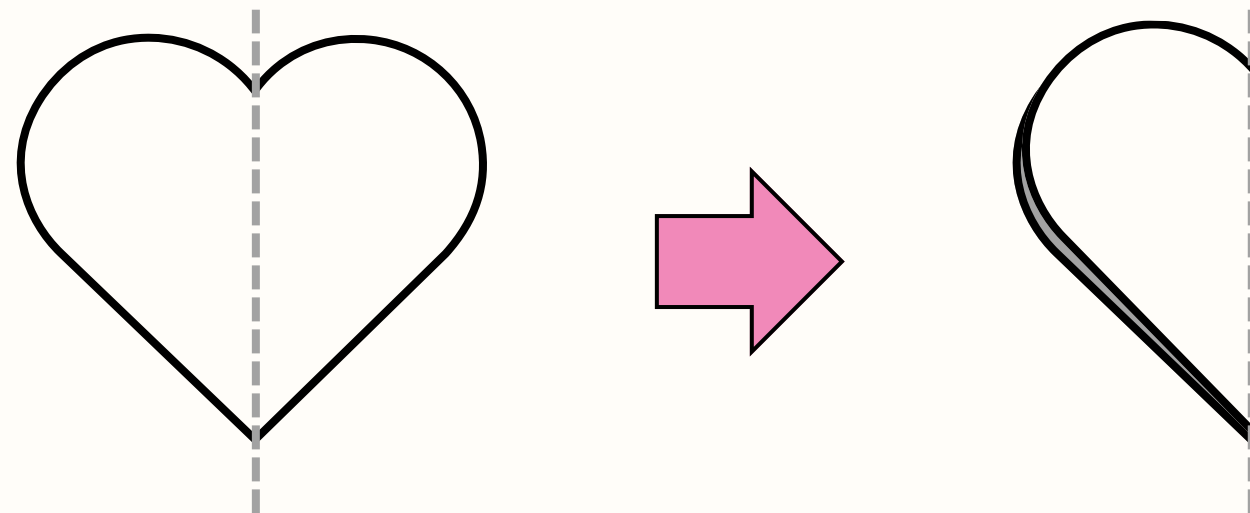
What is Symmetry?

Symmetry is when a shape or object looks exactly the same on both sides.

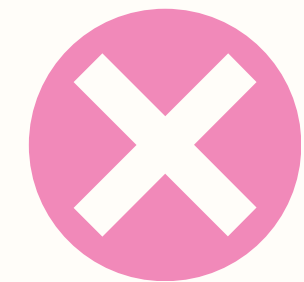
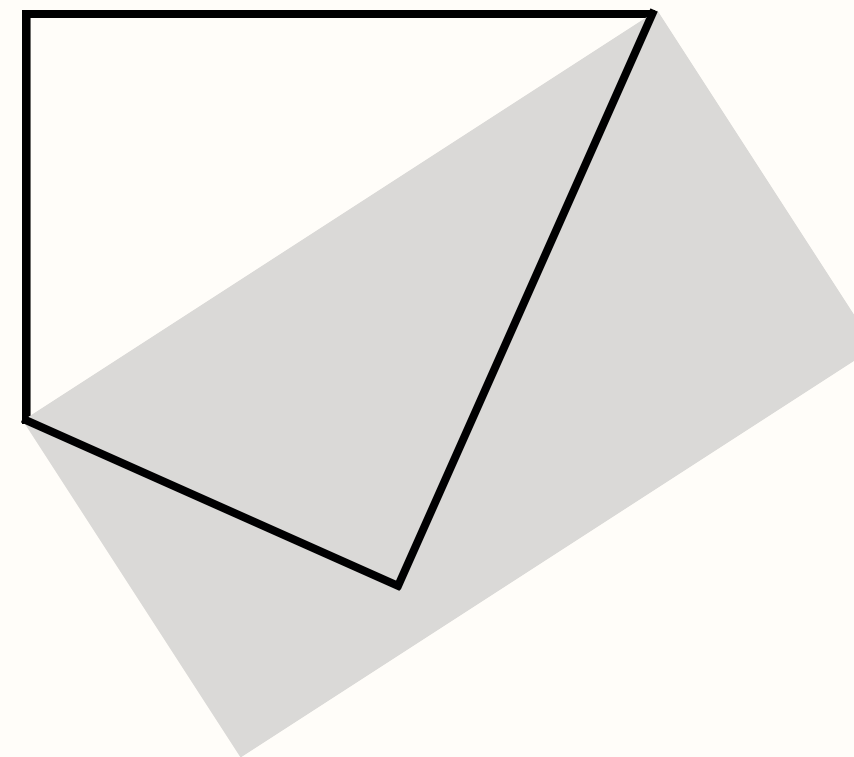
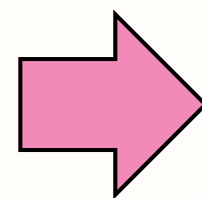
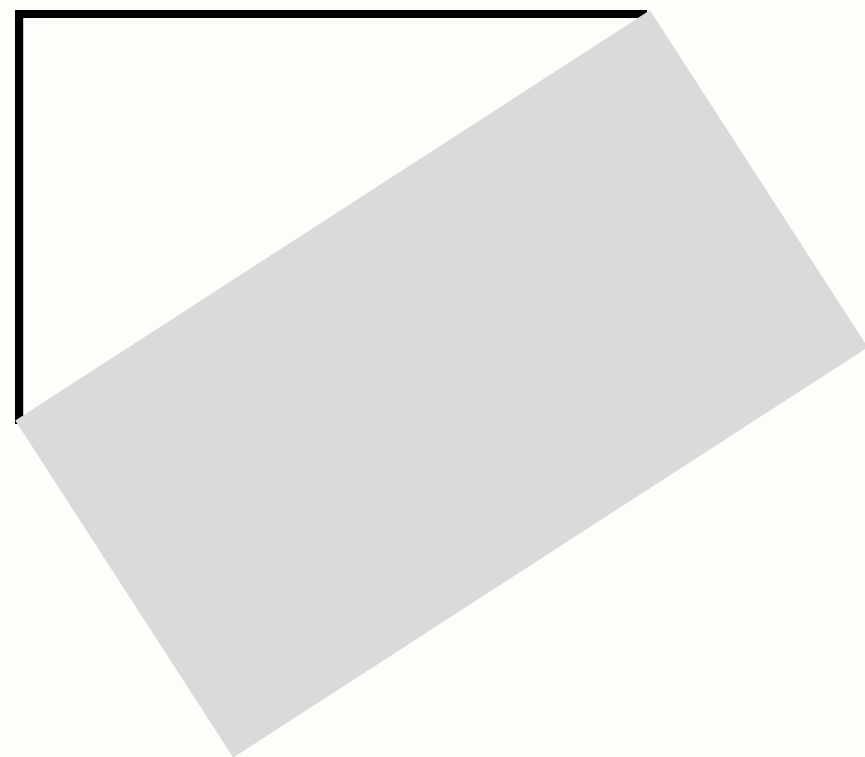
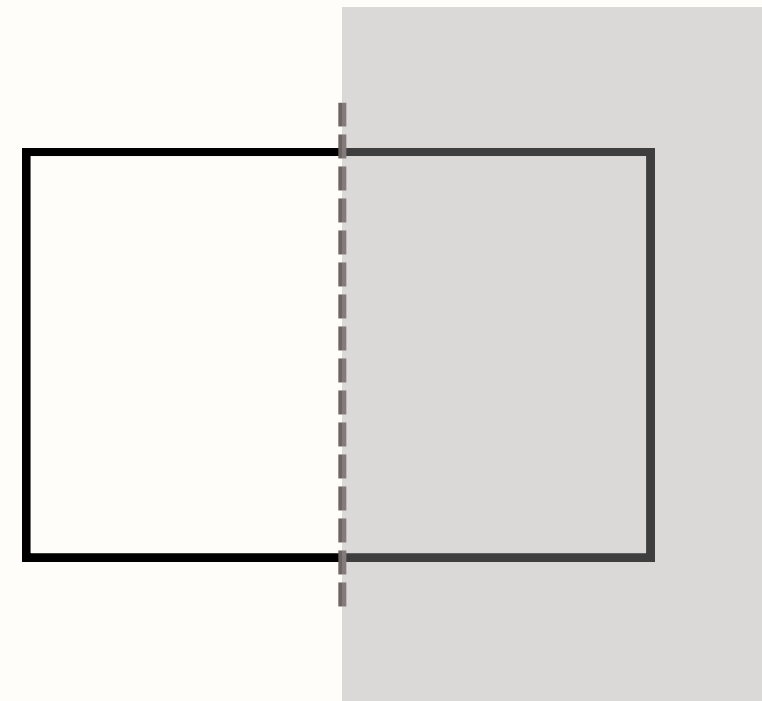
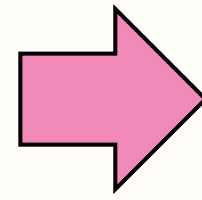
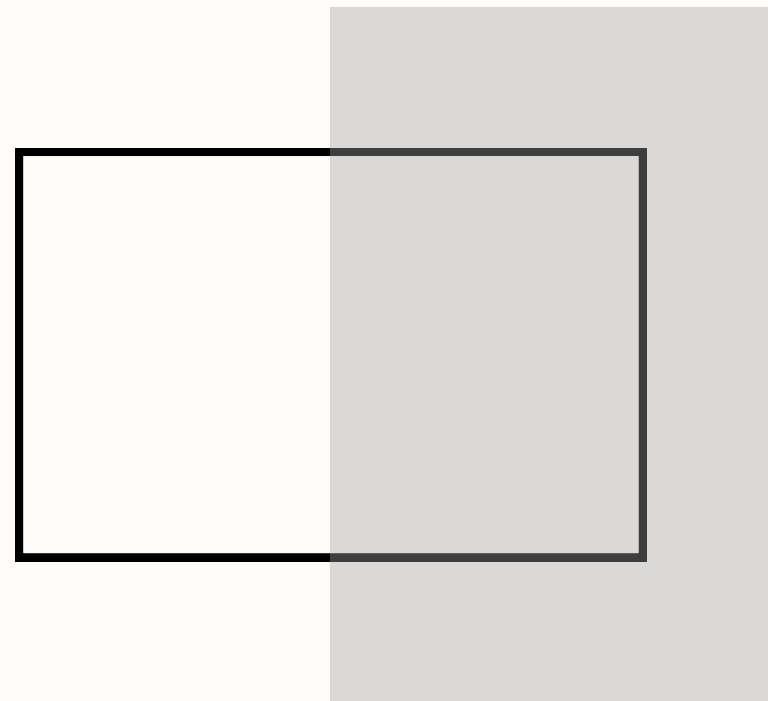


Identifying Lines of Symmetry

A line of symmetry is an imaginary line where you can fold/cut a shape, leaving you with two identical halves.

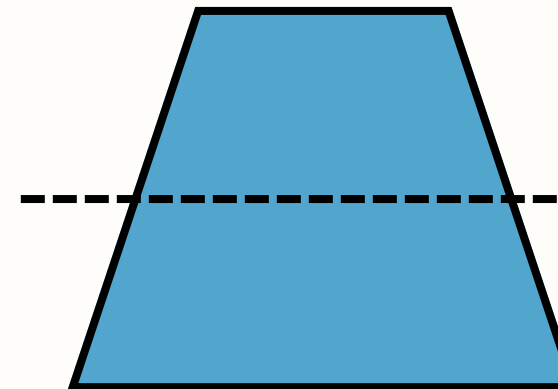
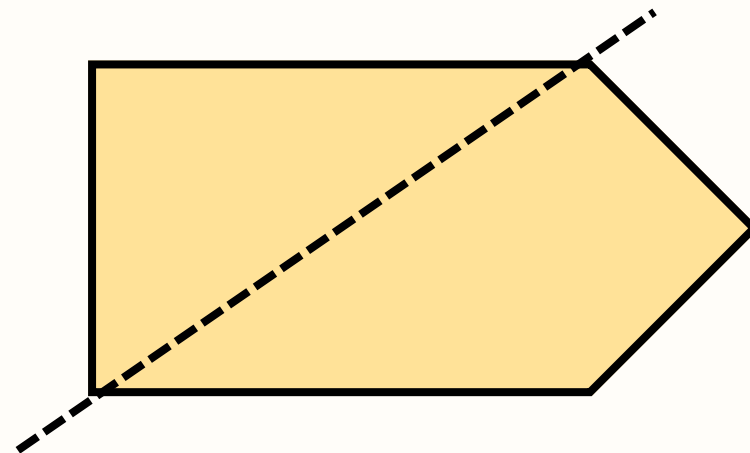
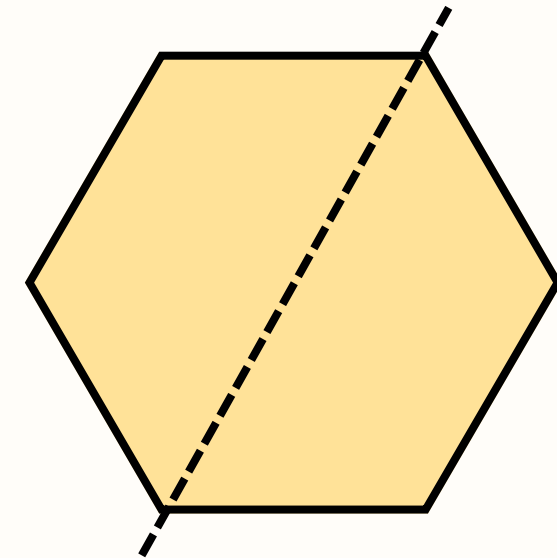
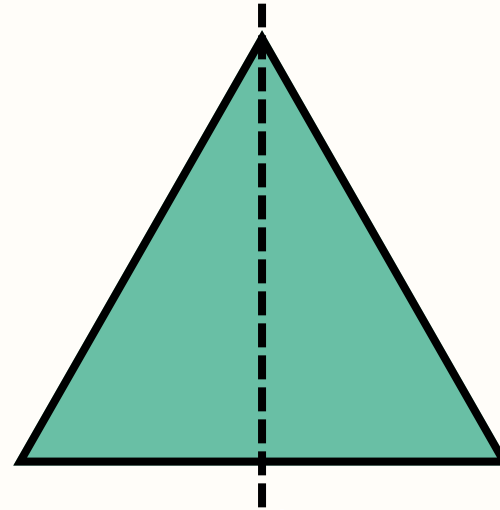
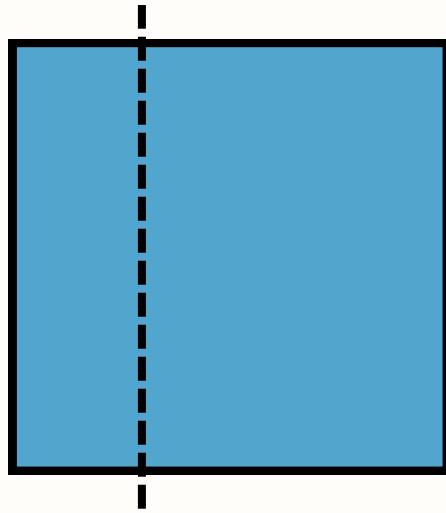


You may also be able to use a mirror to identify lines of symmetry:



*No longer looks
like a rectangle*

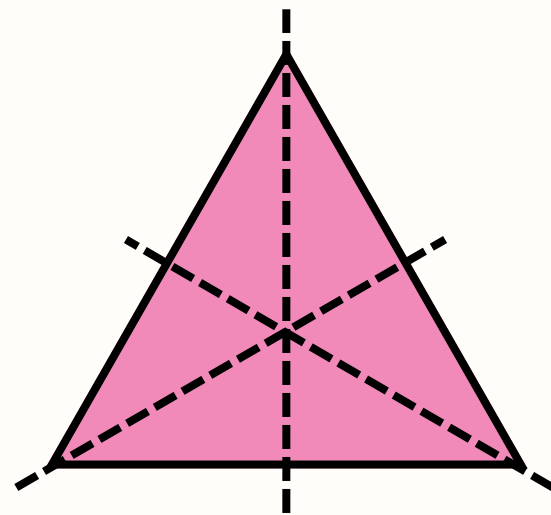
Tick the shapes where the line of symmetry has been identified correctly:



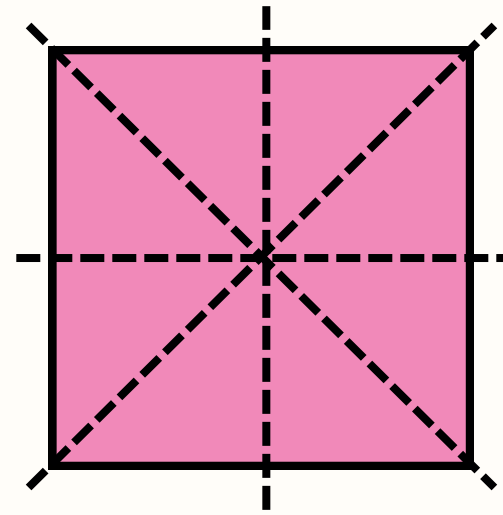
Lines of Symmetry in Regular Shapes

A regular shape has equal side lengths and equal angles.

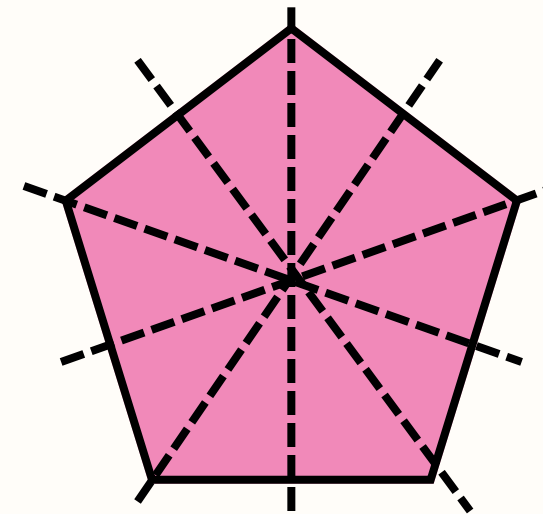
Count along to see how many lines of symmetry the regular shapes have:



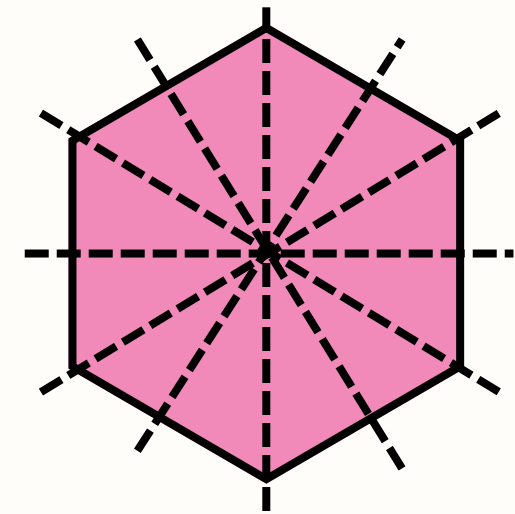
3



4



5



6



THANK YOU

Grade 3

